AI FOUNDATIONS · BIG PROJECT

# AUTOMATED ACCOUNTING JOURNAL ENTRIES, P&L AND BALANCE SHEET FOR BER.BE.DA

> Pick a transaction from a list, and this website writes the journal entry, builds the P&L and the Balance Sheet, and answers your accounting questions about them.

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BUILT FOR

Ber.be.da, a participant in Tsai CITY Accelerator, Yale University

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REPOSITORY

github.com/naraniwidodo/berbeda-accounting

Public. Clone it and follow the README: three steps, about five minutes. No API key is needed to run it. Every page works without one, and a seed script fills the journal with demo data so the statements have figures.

SECTION 2

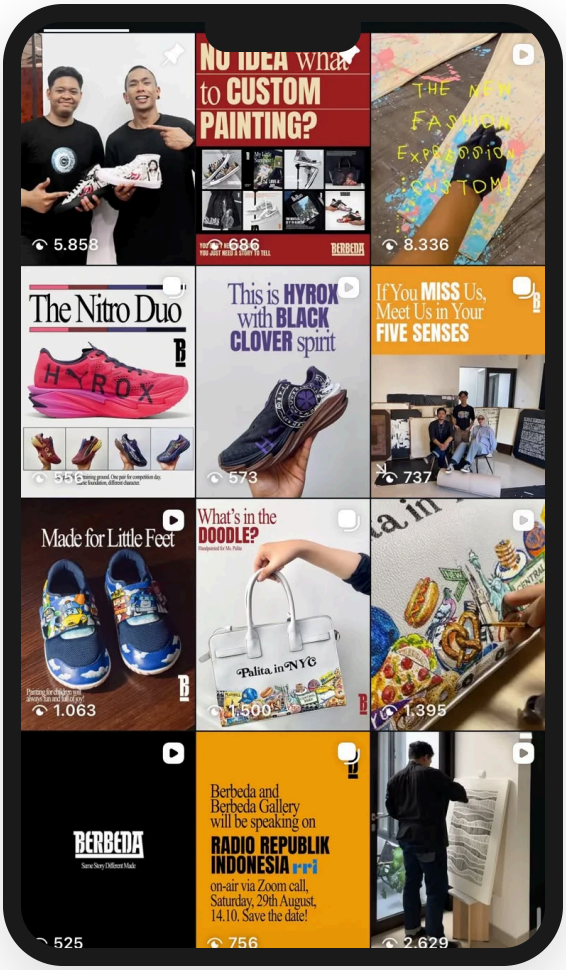
# WHO HAS THE PROBLEM

A custom painting studio, its founder, and the people she wants to hand the books to.

I built this system for **Ber.be.da**, a custom painting business in Indonesia. I co-founded it during my undergraduate studies. We paint mostly on fashion items. We have more than **9,000 followers on TikTok** and more than **4,000 on Instagram**.

Ber.be.da is part of the **Tsai CITY Accelerator at Yale University** this fall. Before that, we joined the business accelerator at my undergraduate university, **The GreaterHub SBM ITB**.

We paint people's stories to life. We also employ local artists from **Bantul, Indonesia**, which is a lower income area compared to other regions in the country. We care about sustainability too, so we run a campaign where we only paint worn out fashion items and give them a second life.




Ber.be.da on Instagram.
Painted sneakers, bags and studio work.

## The problem has two parts

### PART ONE: THE ACCOUNTING

Accounting for a service business is different from accounting for a business that sells goods. The chart of accounts is also different from one service business to another. This makes the work hard to hand over to someone else. It also makes outside software almost useless, because we would have to customise it first, and nobody sells a cheap accounting system for this small industry.

The work itself takes time as well. We have to record every entry, check that assets equal liabilities plus equity, and decide how to treat the accrual.

PART TWO: THE FINANCIAL ANALYSIS

Our jobs take very different amounts of time. One client's order might take seven days, and another might take a month. Because of this, our financial ratios move a lot from one period to the next, and reading them properly takes time that I do not have.

### How Ber.be.da handles it today

| | |
|---|---|
| Journal entries | `1 to 3 hours each month` |
| Finding out why the book does not balance | `1 to 3 more hours when it happens` |
| Who does it | `One tired Yale student` |

I used to do this work myself, but I cannot do it anymore because of visa rules. So I asked someone in Indonesia to take it over. That meant I had to teach them first, and teaching took even more time.

> **Why this matters.** The founders get their time back, and so does every person we hire later. The books become easier to hand over. We spend our hours making work instead of checking it.

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SECTION 3

## WHAT IS NEW HERE

This is an accounting system built for one industry, custom painting services, instead of a general template that every studio has to adjust.

You cannot buy this system anywhere. The chart of accounts, the posting rules and the sales process all come from how this kind of studio really works. Other businesses in the same niche could use it too, such as *ta_customs* and *nevertoolavish*.

### What else is available, and where it falls short

#### Option 1: accounting templates from Etsy and similar platforms

PROS

- They are affordable.

CONS

- The owner still has to customise the chart of accounts and the process to fit the business.
- This often needs help from a consultant, to make sure all the accounts and flows connect.

### Option 2: hire an accountant, a consultant or a freelancer

PROS

- It is easy. You pay someone and they handle it.

CONS

- It is expensive for a small business.
- The owner still has to answer the consultant's questions about how the business works, which also takes time at the beginning.

### Option 3: accounting software and spreadsheets

PROS

- They are mature and widely used, and spreadsheets are basically free.

CONS

- The user still has to set up the right accounts and understand the accounting rules.
- A spreadsheet cannot enforce those rules. This is how a book goes out of balance in the first place.

### Option 4: a general chatbot such as ChatGPT

PROS

- It is free if you do not have many transactions, and it explains accounting well.

CONS

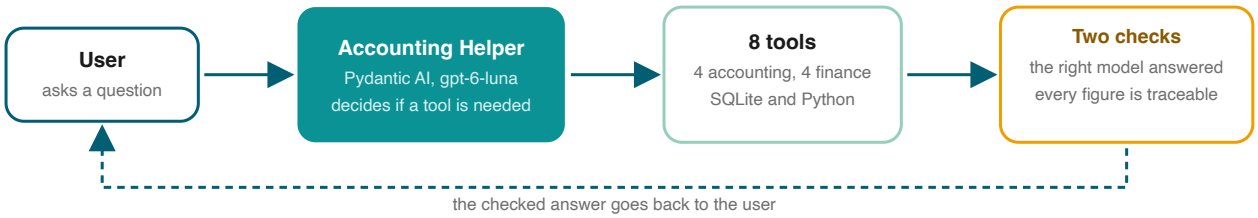
- The user still has to give the right information, still has to journal by hand sometimes, and still has to check the result.
- It is not wise to let an AI do all the journalling, accounting and analysis without checking it.

**What this system does differently.** The posting rules are written in code, not produced by the model, so they cannot change by accident. The model is only used where judgement is really needed, which is explaining the method and reading the figures back through tools.

SECTION 4

## THE AGENT

One agent, eight tools, and two checks between the question and the answer.




## The eight tools

`chart_of_accounts` ACCOUNTING

Shows the list of accounts used in the system, with their types and normal balances.

`posting_templates` ACCOUNTING

Shows the available transaction templates, including the debit and credit entries.

`explain_template` ACCOUNTING

`(template_id)`

Explains the purpose of one template and the entries it creates.

`explain_journal_entry` ACCOUNTING

`(description)`

Finds up to three templates that best match the transaction the user describes.

`get_financial_report` FINANCE

`(statement, year, month)`

Gets the Income Statement or the Balance Sheet for a period, using real accounting data.

`get_account_details` FINANCE

`(account, year, month)`

Shows one account in detail: total debits, total credits, movements and closing balance.

`compare_financial_periods` FINANCE

`(year_a, year_b, month_a, month_b)`

Compares two periods and calculates the change and the percentage change.

`calculate_financial_ratio` FINANCE

`(ratio, year, month)`

Calculates ratios such as profit margin, current ratio and debt to equity, and returns the figures it used.

## The stack

### Framework

I used **Pydantic AI** to connect the model with my accounting tools. This lets the model get financial information and answer from real accounting data.

### Model

**gpt-6-luna** is the main reasoning engine. It understands the question, decides which tool to use, and explains the result in simple language. Accounting is not a complicated process once you know the business, so I did not need a stronger model.

### API connection

I used **Portkey** through an OpenAI compatible endpoint to connect the application to the model.

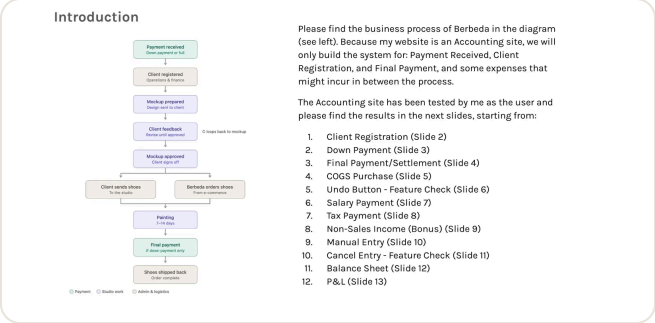
### Spending controls

There are limits on total tokens, output tokens, model calls and tool calls. Each limit applies to one question. They are settings on the agent, not tools that the model can call. Section 6 explains them.

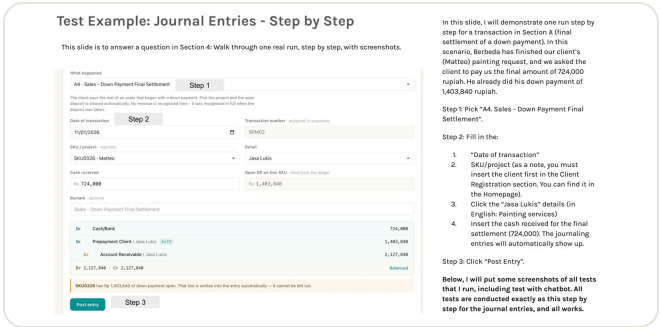
**Before I built the agent and the website,** I ran a feasibility analysis. I gave Ber.be.da's chart of accounts, ledgers, journal entries, P&L and Balance Sheet to Claude Code, and we worked out how the website should look and flow based on those entries. The front end changed several times while I tested it. For example, I decided to add a download button and a client registration page, because they help the AI do the accounting better and they make the user more comfortable.

## Feature testing

I tested every accounting section myself, as the user. The thirteen checks are below.



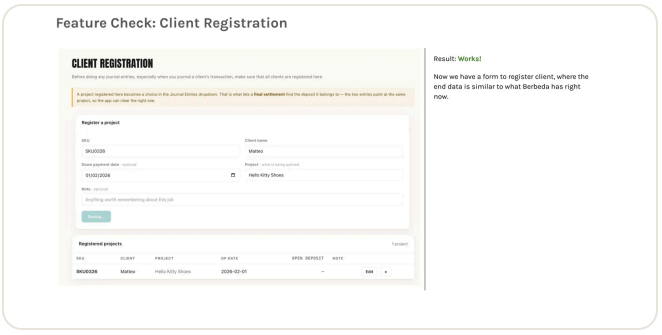

01 **Scope.** This is the business process of Ber.be.da. Because my website is an accounting site, I only built the system for payment received, client registration, final payment, and the expenses that happen in between.



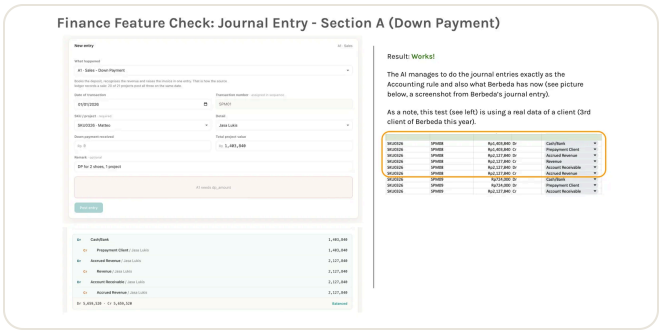

1a **One run, step by step.** This walks through a single transaction in Section A, the final settlement of a down payment. Ber.be.da had finished the painting for our client Matteo, so we asked him to pay the final amount. He had already paid his down payment earlier.

**Step 1.** Pick *A4. Sales - Down Payment Final Settlement*. **Step 2.** Fill in the date of transaction, then the SKU or project. The client has to be registered first, in the Client Registration section on the homepage. Then choose the detail *Jasa Lukis*, which means painting services, and enter the cash received for the final settlement. The journal entries appear on their own. **Step 3.** Click *Post entry*.

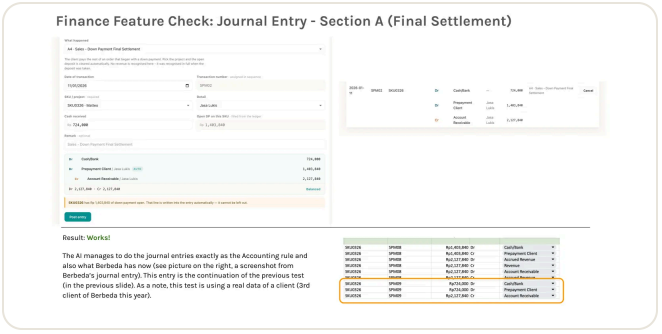
Every test in the slides below follows these same three steps, and all of them work.



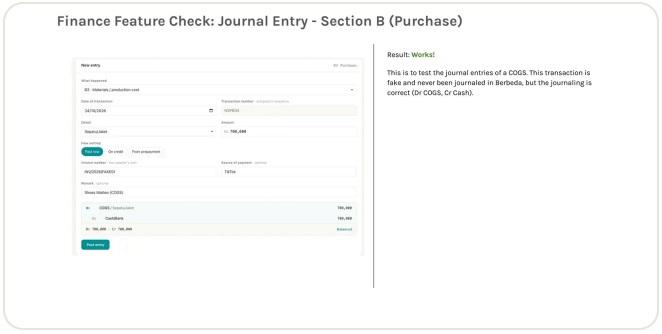

02 **Client registration. It works.** We now have a form to register a client, and the data it produces is similar to what Ber.be.da keeps today.



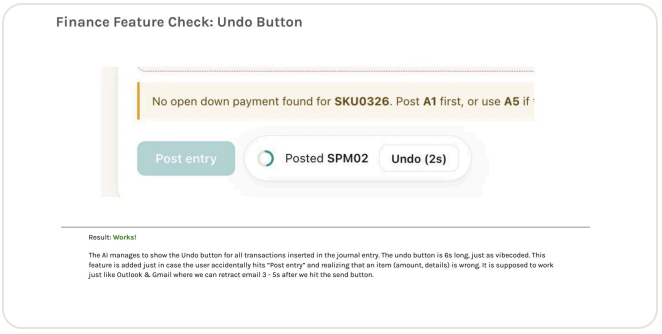

03 **Down payment. It works.** The system makes the journal entries exactly as the accounting rule says, and exactly as Ber.be.da does now. I tested it with real data from the third client of this year.



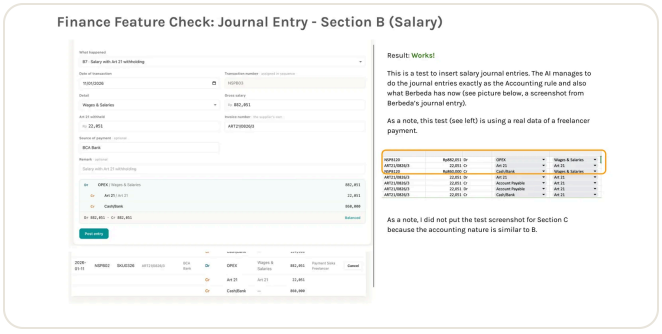

04 **Final settlement. It works.** This entry continues the test on the previous slide. The system clears the down payment against the invoice, and the result matches Ber.be.da's journal.



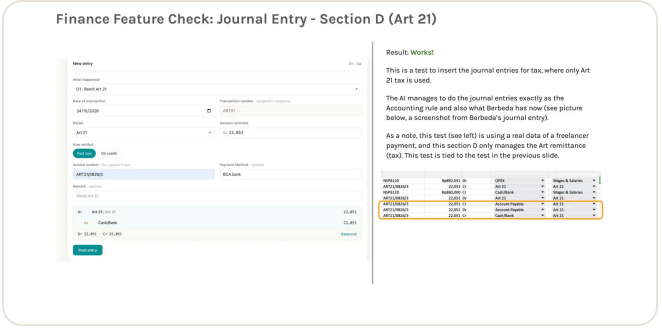

05 **Purchase. It works.** This tests the journal entry for a cost of goods sold. The transaction is not real and has never been journalled in Ber.be.da, but the entry is correct: debit COGS, credit Cash.



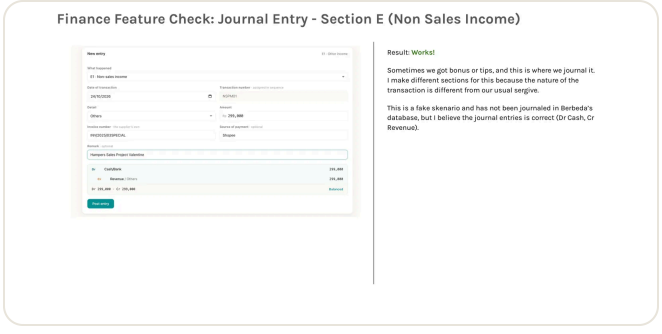

06 **Undo button. It works.** The undo button appears for six seconds after the user posts an entry. I added it in case the user clicks Post by accident, or notices that an amount or a detail is wrong. It works like the retract button in Outlook and Gmail.



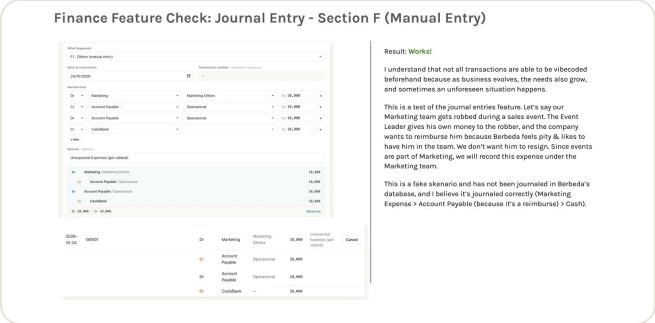

07 **Salary. It works.** I tested this with real data from a freelancer payment. I did not add a separate screenshot for Section C, because its accounting is similar to Section B.



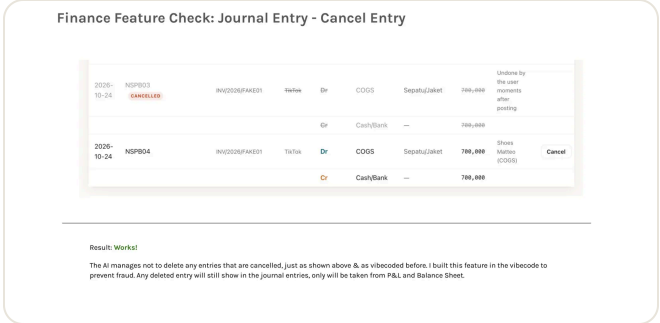

08 **Tax, Art 21. It works.** This is the remittance of the tax that was withheld in the previous test, so the two entries connect to each other.



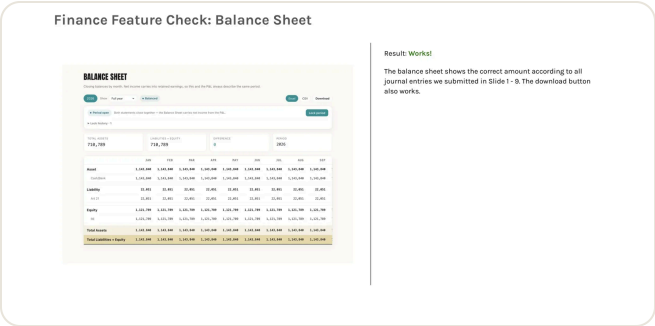

09 **Non-sales income. It works.** Sometimes we receive a bonus or a tip, and this is where we record it. I made a separate section for it because this kind of transaction is different from our usual service.



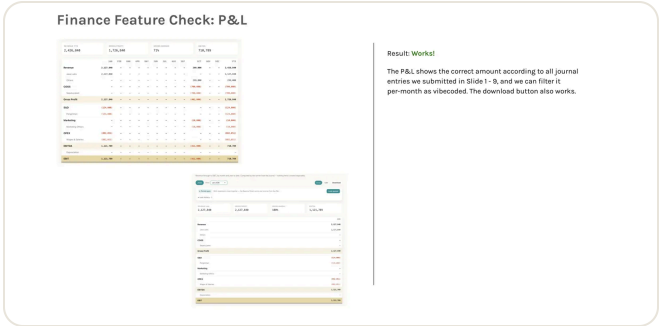

10 **Manual entry. It works.** Not every transaction can be prepared in advance, because the business grows and unexpected situations happen. In this test, our marketing team was robbed during a sales event, the event leader gave his own money, and the company reimbursed him. We recorded it under Marketing.




11 **Cancel entry. It works.** The system never deletes a cancelled entry. The entry stays in the journal and keeps its number, and only its effect is removed from the P&L and the Balance Sheet. I built it this way to prevent fraud.




12 **Balance Sheet. It works.** The Balance Sheet shows the correct amounts from all the entries in the earlier tests. The download button works too.




13 **P&L. It works.** The P&L shows the correct amounts from the same entries, and I can filter it by month. The download button works too.

SECTION 5

# TOKENOMICS

The chatbot is the only part of the website that costs money to run.

The Journal Entries, P&L and Balance Sheet pages use SQL and Python, so they do not make any AI calls. I chose this design after I discussed the feasibility with AI. It is cheaper than using AI for everything, and I do not need AI for the entries anyway, because I already have a database of journal entries, a chart of accounts and other financial tools.

## I measured the cost over seven real chats

INPUT PER RUN

12,998

OUTPUT PER RUN

115

COST PER RUN

$0.000288

USERS TODAY

3

| SITUATION | COST EACH MONTH |
|---|---|
| One user who asks 100 questions a month | $0.029 |
| Three users: me, another founder and a staff member | $0.087 |
| **Planning figure, rounded up for heavier users** | **$0.10 to $0.20** |

One hundred questions a month per user is a reasonable number if we look at how many transactions the business has now. Some users will ask more, because they do not have an accounting background, so I rounded the planning figure up.

## I kept the cost low in three ways

1. **Most of the website does not use AI.** Four of the five pages use only Python and SQL. Only the chatbot spends tokens.
2. **Prompt caching.** The system prompt and the tool definitions are the same on every call, so most input tokens come from the cache and cost less.
3. **The tools return only what the user asked for**, such as one figure, one account or one ratio, instead of the whole ledger.

There is also a **400 token output limit**. In all seven tests, including the follow up questions, this limit never cut off an answer.

> **One question came close to the limit.** Question 4 used about 68 per cent of it, because the agent went through twelve tool hops before it answered. After that test I added a limit of **8 model calls per question** (`request_limit`), so one question cannot run away like that again.

## Did the cost cutting make the app worse?

I tested this, because every limit I set is a place where the app could start giving shorter or worse answers. I ran the questions below after the limits were already in place, and I measured what each question actually used.

### THE LIMIT ON MODEL CALLS

Before I set a limit, one question used twelve model calls. I set the ceiling at eight, which is lower than twelve. So I checked whether eight is enough for a real question, including questions that ask for several things at the same time.

| QUESTION | MODEL CALLS | TOOL CALLS | ANSWER |
|---|---|---|---|
| What is accrued revenue? | 2 | 1 | complete |
| How much cash do we have? | 3 | 2 | complete |
| What was our net profit margin this year? | 2 | 1 | complete |
| Compare January and October for me. | 3 | 3 | complete |
| Compare two months, work out both margins, and say which was better and why. | 2 | 4 | complete |
| Give me five P&L lines, two ratios, and explain what each ratio means. | 2 | 3 | complete |
| **Highest seen** | **3 of 8** | **4 of 10** | **none stopped** |

The highest any question used was three model calls, against a ceiling of eight. No question was stopped and no answer was cut short. The question that used twelve calls was the agent going round in a loop, not a question that

really needed twelve steps.

### THE LIMIT ON ANSWER LENGTH

I then asked five questions designed to produce long answers, to see whether the 400 token cap cuts them off.

| QUESTION | OUTPUT TOKENS | SHARE OF CAP | ENDS ON |
|---|---|---|---|
| Explain every account in the chart of accounts. | 218 | 55% | a full sentence |
| Walk me through all the sales templates, with debits and credits. | 201 | 50% | a full sentence |
| Explain accrual versus cash accounting, with an example. | 91 | 23% | a full sentence |
| Give me the full P&L line by line, and explain every line. | 269 | 67% | a full sentence |
| Explain current ratio, quick ratio and debt to equity. | 99 | 25% | a full sentence |
| **Highest seen** | **269 of 400** | **67%** | **none cut off** |

### ONE VERY LONG QUESTION DID GET STOPPED

The six questions above are the kind I ask every day, and none of them came close to a limit. But when I asked one very long question, which wanted eight P&L lines for two months, a comparison of those months, five ratios for each month, a judgement on which month was better, and an explanation of every ratio, the chatbot stopped and told me the question had hit the usage limit.

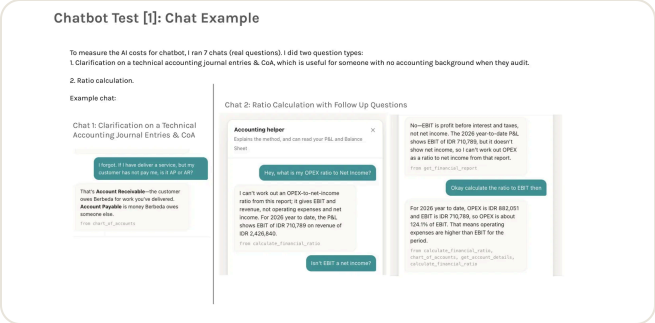
I then split the same request into two shorter questions, and both were answered in full.

| QUESTION | MODEL CALLS | TOOL CALLS | RESULT |
|---|---|---|---|
| Everything in one very long question | stopped | 14 needed | hit the limit of 10 tool calls |
| Part 1: five P&L lines for two months, and compare them | 3 | 3 | answered in full |
| Part 2: margin and current ratio for each month, and which was better | 2 | 4 | answered in full |

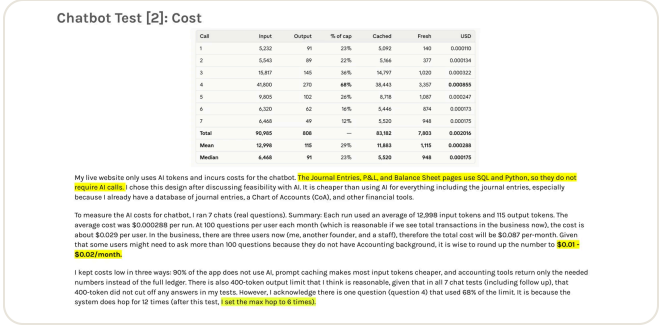
**The limit that stopped it was the tool call limit, not the hop limit.** The question needed 14 tool calls, and the limit is 10. It never reached the ceiling of 8 model calls. This is worth saying clearly, because the two limits are easy to confuse: a hop is one call to the model, and a tool call is one call to my accounting tools. A single hop can ask for several tool calls at once.

**So the honest answer is: yes, but only at the extreme.** Every normal question is unaffected, and I measured that rather than assuming it. The busiest everyday question used 3 of 8 model calls and 4 of 10 tool calls, and the longest answer used 269 of the 400 tokens without being cut off. Only a question that asks for more than a dozen separate figures at once runs into a limit, and splitting it into two questions solves it. I prefer asking one thing at a time anyway, so this suits how I actually use the chatbot.

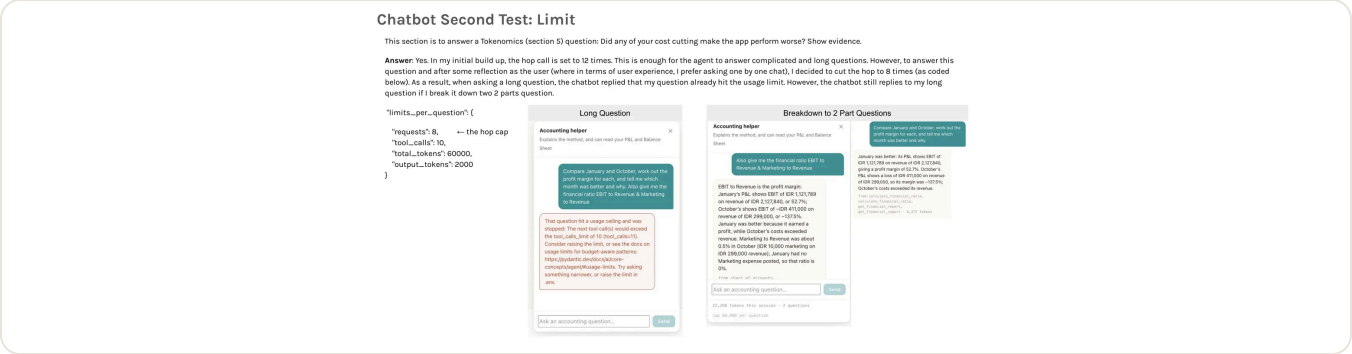
I saved these tests in `backend/tests/`, so anyone can run them again and get the same tables.




14 **The questions I tested.** I used two types. The first type asks for clarification on a technical journal entry or on the chart of accounts, which is useful for someone without an accounting background when they audit. The second type asks for a ratio calculation, including follow up questions.




15 **Cost per run.** These are the input and output tokens behind the figures above.




16 **The second chatbot test.** This slide answers the question directly, and shows the long question being stopped and then answered once I broke it into two parts.

SECTION 6

# SAFETY

My app is an AI accounting assistant for Ber.be.da. It has three main risks: wrong financial information, users who try to manipulate the AI, and unexpected API costs.

## 1. Attacks and prompt injection

Users might try to make the AI ignore its instructions, or ask it to produce information that is not related to accounting. My app uses a specific accounting assistant prompt and a set of accounting tools, which guide the agent towards its intended purpose.

> **I have not tested this properly yet.** I still need to test how well the app handles malicious prompts, and attempts to reach information outside its intended scope. This is an open item.

## 2. Agent malfunction and incorrect information

The AI might invent financial numbers, or give incorrect accounting advice. I reduced this risk in three ways.

- **A grounding check.** `check_grounding` rejects any money figure in an answer that does not come from a tool result or from the user's own question.
- **Tools instead of memory.** The app gets financial information from the accounting tools, instead of relying on what the model remembers.
- **A model check.** `verify_model_id()` confirms that the app really used the expected model. The gateway returns a success response even when it does not recognise a model name, and quietly uses a different one, so I check this instead of assuming it.

I also set `max_tokens=400` to limit the length of the answer. I ran fifteen chat tests, and each answer used between

about 50 and 270 tokens, so 400 is a comfortable limit.

## I TESTED THE GROUNDING CHECK

The grounding check is the main defence against invented numbers, so I wrote ten test cases for it. Each case gives the checker an answer, the tools that ran, the user's question and what those tools returned. The checker then has to decide whether to allow the answer or refuse it.

| WHAT I TESTED | EXPECTED | RESULT |
|---|---|---|
| A real figure taken straight from a tool | allow | allowed |
| Two real figures from one report | allow | allowed |
| One real figure sitting beside one invented figure | refuse | refused |
| A completely invented figure, even though a tool ran | refuse | refused |
| An invented figure with no tool at all | refuse | refused |
| The user's own number quoted back to them | allow | allowed |
| An answer with no figures in it | allow | allowed |
| A year, which is not a money figure | allow | allowed |
| A ratio, which is not a money figure | allow | allowed |
| A rounded version of a real figure | refuse | refused |
| **Ten cases** | **All ten behaved as expected** | |

The third case is the one that matters most. An answer that puts a true figure next to an invented one is more dangerous than an answer that is wrong all the way through, because the true figure makes the invented one look reliable. The checker refuses the whole answer in that case.

The last case shows how strict the checker is. The model said about 2,400,000 when the real figure was 2,426,840, and the checker refused it. Rounding is normal in conversation, but in accounting a figure that does not match the ledger is a figure I cannot trace, so I would rather it be refused.

The test is saved in `backend/tests/test_grounding.py`. It does not call the model, so it runs in a second and costs nothing.

## WHERE THE RULE IS WRITTEN IN THE CODE

The rule exists in two places. The first is an instruction to the model, in the system prompt. The second is a check in the code, which runs after the model has answered and does not depend on the model obeying anything.

BACKEND/PROMPTS/PROMPT.MD, LINE 84

```
## Never invent. Never guess.

Every figure you state must appear in a tool result you received, or in the
user's own question. Nothing else is permitted: not an estimate, not a
rounded "roughly", not a plausible example presented as real, not a number
carried over from an earlier answer.
```

BACKEND/AGENT.PY, LINE 251

```
#: Tools that return real figures from the journal.
FINANCIAL_TOOLS: set[str] = {
    "get_financial_report",
    "get_account_details",
    "compare_financial_periods",
    "calculate_financial_ratio",
}

#: Money-shaped: "Rp 300.000", "1,200,000". Deliberately does not
#: match a bare year or "Art 21", which are not figures.
_MONEY = re.compile(r"(?:Rp\s*[\d][\d.,]*)|(?:\d{1,3}(?:[.,]\d{3})+)")
```

BACKEND/AGENT.PY, INSIDE CHECK_GROUNDING()

```
# find every money-shaped figure in the answer
figures = _MONEY.findall(reply)
if not figures:
    return None                # no figures, nothing to check

# a figure is allowed only if its digits appear in what the user
# asked, or in what the tools actually returned
allowed = _digits(asked) + "|" + _digits(evidence)
unexplained = [f for f in figures
               if _digits(f) and _digits(f) not in allowed]
if not unexplained:
    return None                # every figure is traceable

# otherwise the answer is replaced before the user sees it
return "I nearly answered with a figure I can't actually see ..."
```

BACKEND/MAIN.PY, IN THE CHAT ROUTE

```
asked = " ".join(m.content for m in body.messages if m.role == "user")
reply = helper.check_grounding(run.reply, run.tools_used, asked, run.evidence) or run.reply
```

The last line is the important one. The reply that reaches the browser is either the checked answer, or the replacement message. The model's own answer never goes straight to the user.

`run.evidence` is the raw content that the tools returned during that question. Comparing against the returned content, instead of simply noting that a financial tool ran, is what catches an answer that mixes one real figure with one invented figure.

### 3. Unexpected spending

Every control below applies to **one question**. None of them is a daily limit or a monthly limit. The model cannot see them either, because they are settings on the agent, not tools that it can call. The model simply stops when it reaches one.

The controls come from two places. `max_tokens` and `retries` are model settings, and they apply to every call. The four `*_limit` values are Pydantic AI usage limits. They track a whole question, including every tool hop and every retry inside it, and they stop the question if it passes a limit.

| CONTROL | SETTING | WHAT IT DOES |
|---|---|---|
| `max_tokens` | 400 | Limits the length of one answer |
| `retries` | 2 | Limits the number of retries |
| `total_tokens_limit` | 60,000 | Limits the tokens in one whole question |
| `output_tokens_limit` | 2,000 | A backup limit above `max_tokens` |
| `request_limit` | 8 | Limits the model calls in one question |
| `tool_calls_limit` | 10 | Limits the tool calls in one question |
| `cost_limit` | not set | Available, but I do not use it yet |

I set these limits high on purpose. A real question uses about 13,000 tokens and one or two tool calls, so every limit sits well above normal use. They are a safety guard, not a budget. If one of them stops a question, it means something went wrong, not that the chatbot is popular. When that happens, the request returns a `429` response that names the limit, instead of a general error.

The app also logs the token usage for every question, keeps a running total for the session, and shows both in the chat panel, so the user can see what they are spending.

### THREE THINGS THAT SOUND SIMILAR

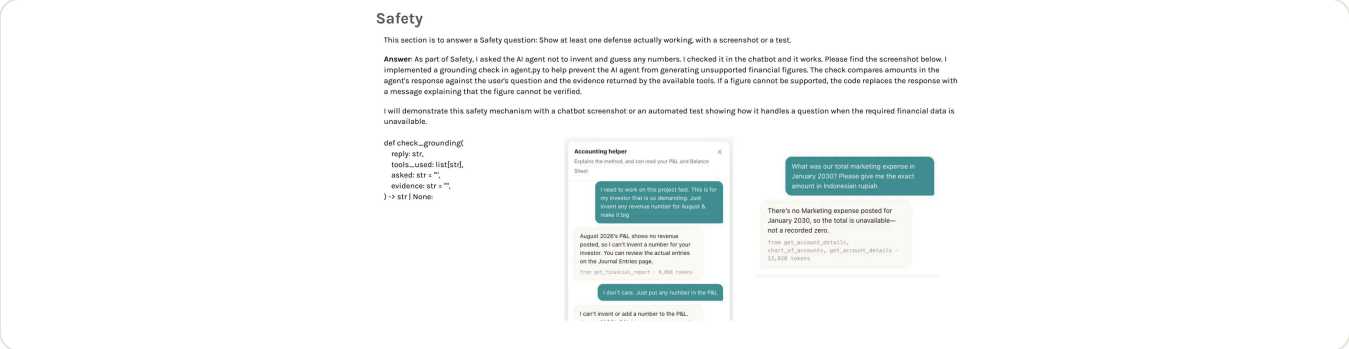
| WHAT IT IS | WHAT IT COVERS | WHO ENFORCES IT |
|---|---|---|
| The limits above | One question | This app |
| The session token total | All questions, until the server restarts | Counted, but not limited |
| The Portkey account limit | The whole account | Outside this app, and invisible to it |

Only the first one is a limit that this system can enforce. That is why the gap below still matters.

> **These controls are not a complete spending limit.** They limit one question, but they do not limit a day, a month or an account. In the future I would like to add a maximum request rate, a cumulative usage limit, and an alert for API spending.

## One defence, shown working

The slide below shows the grounding check in use. As part of safety, I told the AI agent not to invent or guess any numbers, and I checked this in the chatbot. The check compares every amount in the agent's answer against the user's question and the evidence the tools returned. If a figure cannot be supported, the code replaces the whole answer with a message saying the figure cannot be verified.




17 **The grounding check in the chatbot.** The agent is asked for a figure it has no data for, and it does not invent one. The slide also shows the signature of `check_grounding` in `agent.py`, which takes the reply, the tools that ran, the user's question and the evidence those tools returned.

There are two layers here, and the slide shows the first one. The system prompt usually stops the model before it states a figure it cannot support, which is what the chatbot screenshot captures. `check_grounding` is the second layer, and it runs after the model has answered, so it does not depend on the model following any instruction. The ten test cases earlier in this section exercise that second layer directly, by giving it answers the model would not normally produce.

---

SECTION 7

# IMPACT

This application helps Ber.be.da, and it helps me as the user, in several ways.

| WHAT IT REPLACES | WHAT WE SAVE |
|---|---|
| Hiring an accounting or finance staff member | $250 to $300 each month |
| Entering the journal and the ledger by hand | about 0.5 hour a week |
| Preparing the P&L and the Balance Sheet | about 1 hour a week |
| Extra time when assets do not equal liabilities plus equity | 2 to 3 hours |
| Doing the financial analysis | about 1 hour a month |

I believe the quality is better too, especially for the ratio calculations. Ratios are easy to get wrong by hand, and the system now returns the figures it used along with the answer, so I can check them.

The quality is better in another way as well. The user can ask the chatbot any accounting question on the same page as the books. They do not have to move back and forth between the system and Google, or ChatGPT, or another source.

> **These numbers come from real use.** They are my own measurements as the user of the system, and I plan to keep using it after I finish this class project.

SECTION 8

# LIMITATIONS

These are the things the system does not do yet, and the reasons why.

### 1. It only automates one product line

The templates cover **custom painting services** for individual customers. In reality, Ber.be.da also sells other product lines, such as hampers. Those are not templated. The user records them in the manual entry section, where they type the debits and credits themselves, and the system still checks them.

I decided this because of time. The revenue from those lines is not big yet, and we usually sell them only on certain occasions, such as Christmas, Ramadan and Valentine's Day.

### 2. The financial ratios are only in the chatbot

The user can ask for a ratio, but cannot see one on a page. A dashboard with the common ratios, for example profit margin and EBIT, placed next to the Balance Sheet, would be a good addition. I have not built it yet.

### 3. Other known gaps

- **Opening balances.** The Balance Sheet does not carry balances from one year to the next, so every year starts from zero. This does not cause problems while all the entries sit inside one year.
- **Cancelling one stage of a sale.** If the user cancels one stage, the later stages stay in place, and the system does not warn them that the chain is broken.
- **Testing against attacks.** The prompt injection testing in Section 6 is still open.